Information-mediated action at a distance in traffic risk propagation

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Abstract

Introduction

Future automated driving and intelligent transportation systems are evolving from traffic operation paradigms dominated by local perception and vehicle dynamics into complex dynamical systems in which physical motion, information exchange, and intelligent decision-making are tightly coupled. In conventional traffic processes, local braking, speed perturbations, and localized congestion can propagate continuously through vehicle-following interactions, vehicle motion, and traffic waves along vehicle queues and road networks, allowing local disturbances to develop into traffic impacts over larger spatial and temporal scales[1][2]. Meanwhile, vehicle–infrastructure cooperation and cloud-based intelligence are reshaping how vehicles acquire and use traffic information, enabling vehicle decisions to rely not only on the immediate physical environment but also on information from remote vehicles, infrastructure, and broader traffic states[3]. As a result, the formation and evolution of traffic risk in future transportation systems may be jointly shaped by physical and information propagation. This raises a fundamental question: before ordinary physical propagation from a source region begins to affect risk formation in a target region, can risk information originating from that source region alter the risk in another traffic region spatially separated from it?

Existing studies have revealed several mechanisms through which traffic states and risk can evolve across space. At the vehicle scale, risk may progressively accumulate or propagate through successive physical interactions among adjacent vehicles in a traffic stream[4]. Nonlocal traffic-flow models, by contrast, allow vehicles to respond to traffic states over a broader spatial range ahead, indicating that vehicle motion need not depend exclusively on nearest-neighbour information[5][6]. With the development of connected and automated driving technologies, remote information can also enter vehicle control through communication links, while communication delay, link conditions, and dynamic network topology can further affect the stability and safety of closed-loop traffic systems[7][8][3]. Together, these lines of research provide important foundations for understanding physical risk propagation, nonlocal state utilization, and communication-assisted control, but they have not yet established a directly attributable relationship between specific remote risk information and changes in risk within a target region.

Answering this question is not straightforward. A change in risk observed in a distant traffic region does not necessarily imply that the change was caused by remote risk information. Risk-graph and spatiotemporal association approaches can characterize relationships between risk states at different locations[9], but such associations alone cannot identify the independent effect of an information pathway. Nonlocal traffic-flow models typically incorporate remote traffic states directly into vehicle dynamics without explicitly distinguishing information generation, transmission, delivery, and actual adoption. Studies of connected and cooperative control, in turn, have largely focused on how communication conditions affect control performance, although the existence of a communication link, successful message delivery, and an actual behavioural response to the delivered information represent distinct stages. More importantly, even when a source region and a target region are spatially separated, disturbances originating at the source may still influence the target indirectly through multi-step car-following interactions, traffic waves, vehicle motion, or shared traffic conditions. Observing a remote change in risk or the presence of a communication link is therefore insufficient to establish an information-mediated risk effect. The central question of this study is whether, after ordinary physical propagation and other major alternative explanations have been excluded, traceable risk information tied to a specified source-risk event can be adopted by target vehicles and alter their actual behaviour before ordinary physical propagation from the same event does so, and whether the associated incremental change in aggregate target-region risk is already present before ordinary physical propagation begins to influence risk formation in the target region.

We refer to the phenomenon in which traceable risk information from a source event is adopted by target vehicles, alters their actual behaviour, and changes risk in the target region before ordinary physical propagation from the same source event begins to shape risk formation there as *information-mediated action at a distance* in traffic risk propagation. It represents an explicitly information-mediated non-nearest-neighbour risk effect between spatially separated traffic regions. The term does not imply quantum entanglement, superluminal transmission, or any form of unmediated interaction. To test for this effect, we explicitly distinguish the physical traffic network from the information network and use a controlled counterfactual design to manipulate the source-risk treatment and the state of the information channel separately, thereby estimating the incremental effect of specified risk information on aggregate risk in the target region. We further trace the full pathway from the source-risk event through message generation, transmission, delivery, and adoption to actual target-vehicle responses and subsequent changes in regional risk, and test whether the information-induced behavioural response precedes the earliest target-region response attributable to ordinary physical propagation from the same source event.

The study is organised into two stages, Stage I and Stage II. Stage I first identifies the core effect of specified remote risk information and its attribution pathway under physically isolated conditions, thereby excluding ordinary physi-

cal propagation from the source region to the target region as far as possible as a competing explanation. The ordinary physical propagation pathway is then restored, and the same counterfactual design and attribution criteria are applied under physically connected conditions to test whether the information-mediated effect and its underlying pathway persist in the presence of competing physical propagation, and to compare the timing of the earliest actual target-vehicle responses induced by the information pathway and by ordinary physical propagation. Stage I further examines whether the target-region risk effect is already detectable within a pre-specified early evaluation window designed to remain unaffected by ordinary physical propagation from the source event, thereby establishing temporal precedence at the level of the risk outcome. The robustness and limits of the effect are subsequently assessed across different traffic and communication conditions, while real-world trajectory and communication data are used to evaluate the empirical plausibility of the risk measurements and the observability of the communication process. Only if the core Stage I evidence meets the pre-specified criteria does the study proceed to Stage II, which examines whether the identified remote risk information can be incorporated into vehicle control to improve target-region risk outcomes while evaluating the resulting trade-offs among safety-related risk, traffic efficiency, and ride comfort.

Results

Effects of remote risk information under physically isolated conditions

Attribution and pathways of specified remote risk-information effects

Remote risk-information effects and response timing under physically connected conditions

Applicability and failure boundaries of information-mediated action at a distance

Safety-constrained control using remote risk information

Discussion

Methods

Study design and experimental framework

This study investigates risk effects mediated by explicit risk information between two spatially separated local traffic regions. The core experiments focus on longitudinal car-following scenarios, in which the risk considered here specifically refers to collision risk arising from car-following interactions; the corresponding risk measures and source-risk event identification rules are formally defined below. The region in which the source-risk treatment is applied and source-risk events are identified is denoted by $\mathcal{R}^{\mathrm{src}}$, whereas the region in which the effects of remote information are evaluated is denoted by $\mathcal{R}^{\mathrm{tgt}}$. Once a risk event satisfying the pre-specified identification criteria is detected in the source region, traceable risk information tied to that event can be transmitted to the target region through an explicit communication channel and, once adopted by target vehicles, incorporated into their decision-making processes. The primary identification objective is to determine whether such specified risk information can alter the actual behaviour of target vehicles through the information pathway before ordinary physical propagation does so, and whether this behavioural influence subsequently changes the aggregate risk of the target region. The overall study design is therefore organised around the relationships among the source region, the explicit information channel, and the target region, as illustrated in Fig. 1.

Within this framework, conditions in the source region may influence the target region through two distinct pathways. Ordinary physical propagation operates through actual traffic processes, including vehicle motion, car-following interactions, merging interactions, conflict interactions, and traffic waves, and can progressively reach the target region through successive vehicle interactions or connected roadway structures. Explicit information propagation, by contrast, transmits information tied to the source-risk event to target vehicles through a communication channel and, once adopted, allows that information to enter their decision-making processes. Because the two pathways may coexist within the same traffic system, spatial separation between the source and target regions alone does not eliminate the possibility of a physical causal connection between them. We therefore treat ordinary physical propagation and explicit information propagation as distinct pathways of influence and, in the subsequent experiments, control physical connectivity and information-channel state separately to identify the independent effect of specified remote risk information.

Three complementary forms of evidence are required to support *information-mediated action at a distance* in traffic-risk propagation (Fig. 1c). The first is **Effect**: relative to the corresponding counterfactual conditions, specified remote risk information must produce an incremental difference in the aggregate risk of the target region. The second is **Attribution**: this difference must be traceable through the sequence from the source-risk event to message generation, transmission and delivery, target-vehicle adoption, actual vehicle response, and the resulting change in target-region risk, while ruling out explanations arising solely from opening the communication channel, shared traffic conditions, or other major alternative mechanisms. The third is **Temporal precedence**. At the behavioural level, the actual target-vehicle response induced by the specified information must occur before the earliest attributable response in the target region caused by ordinary physical propagation from the same source-risk event. At the risk-outcome level, the target-region risk effect associated with the specified information must also emerge within a pre-specified early evaluation window that remains unaffected by ordinary physical propagation from the source. The behavioural criterion therefore asks whether the information pathway changes target-vehicle behaviour earlier than the physical pathway, whereas the risk-outcome criterion asks whether the regional risk effect is already present before ordinary physical propagation begins to influence risk formation in the target region. Effect, Attribution, and Temporal precedence thus serve distinct evidential roles, and no single criterion is sufficient on its own to support the phenomenon

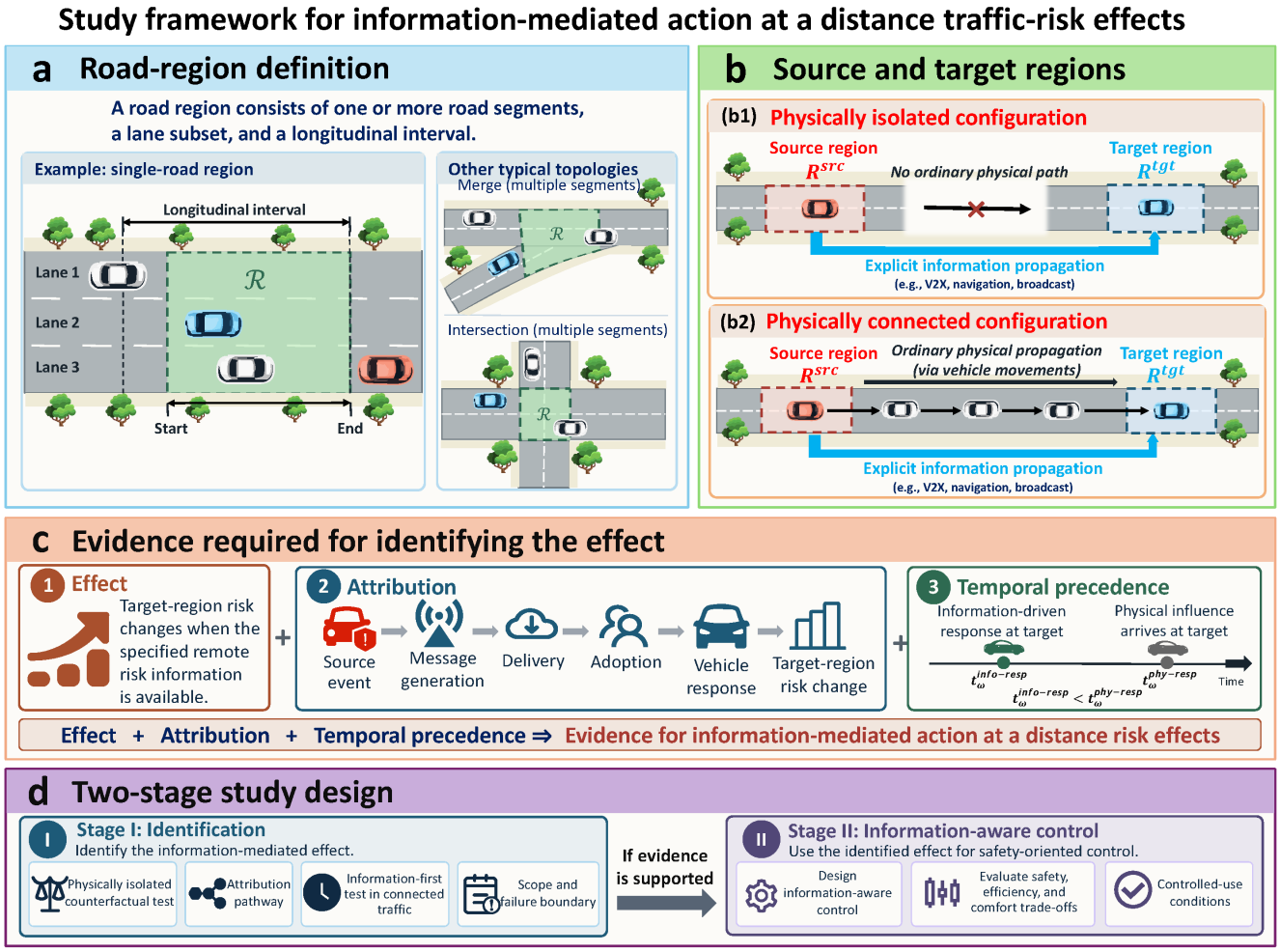


Figure 1: Overall study framework. The figure illustrates the spatial definitions of the source and target regions, the distinction between ordinary physical propagation and explicit information propagation, the three forms of evidence required to identify information-mediated action at a distance in traffic-risk propagation, and the two-stage Stage I–Stage II study design.

defined here as information-mediated action at a distance.

On the basis of these evidential requirements, the study adopts a two-stage design (Fig. 1d). Stage I is designed to determine whether specified remote risk information produces an independent and attributable risk effect that precedes ordinary physical propagation, rather than to optimise control performance. The basic experimental unit in Stage I, denoted by ω, consists of a complete set of four matched counterfactual runs conducted under the same pre-specified traffic scenario and stochastic conditions; thus, each ω represents a matched set of counterfactual runs rather than a single simulation run. The corresponding treatment factors and four experimental conditions are formally defined below. Within each pre-specified scenario, all experimental conditions and baseline control logic other than the manipulated factors are held fixed according to the pre-specified protocol. The study first estimates the incremental effect of remote risk information on target-region risk under physically isolated conditions and tests whether this effect can be attributed to the adoption of the specified information and the resulting actual vehicle responses, thereby excluding ordinary physical propagation as far as possible as a competing explanation. The physical propagation pathway from the source region to the target region is then restored. Under these physically connected conditions, the same counterfactual factors, risk measures, and attribution criteria are retained to test whether the remote information effect and its associated pathway persist in the presence of competing physical propagation, and, within the $(1, 1)$ condition, to compare the timing of the actual target-region responses induced by the information and physical pathways. In parallel, the regional risk interaction effect is re-estimated within a pre-specified early risk-evaluation window shared by all four runs to test whether the risk effect is already present before ordinary physical propagation from the source begins to influence risk formation in the target region. The physically isolated setting therefore provides the core identification of Effect and Attribution, whereas the physically connected setting enables Effect, Attribution, and Temporal precedence to be jointly evaluated within the same experimental framework and further allows their applicability and failure boundaries to be examined across different traffic and communication conditions. Only if the core Stage I evidence satisfies the pre-specified criteria does the study proceed to Stage II, which examines how remote risk information can be incorporated into vehicle control and evaluates the resulting trade-offs among target-region risk, traffic efficiency, and ride comfort under pre-specified safety constraints. Stage II therefore assesses the practical utility of the identified infor-

mation effect, but its control performance does not substitute for the Stage I identification of the effect and its attribution.

Real-world traffic data and controlled simulation serve distinct but complementary evidential roles in this study. Real-world trajectory data are used to calibrate the risk measures and plausible ranges of vehicle motion, and to verify that the simulated traffic states and parameter ranges remain within empirically supported bounds. Real-world communication data are used instead to determine which communication states can be observed directly and which key stages of the information-action pathway remain unobservable. Because the real-world datasets used here do not jointly capture source-risk events, complete message lifecycles, information adoption, actual vehicle responses, and subsequent changes in target-region risk, they are not used to identify the complete information-mediated causal chain directly. Controlled traffic simulation is therefore used to construct counterfactual conditions that are difficult to obtain from observational data, including source-risk treatments, opening and closing of the information channel, and physically isolated and physically connected settings. These controlled conditions provide the basis for Stage I effect identification and temporal-precedence testing, as well as the subsequent Stage II control experiments. Accordingly, real-world data primarily support measurement calibration, empirical range constraints, and observability assessment, whereas the core counterfactual identification and control experiments are conducted in the controlled simulation environment.

Data sources and simulation environment

This study uses three complementary real-world traffic datasets together with a controllable microscopic traffic-simulation platform. The NGSIM vehicle-trajectory data[10][11] are used to support the calibration of longitudinal car-following risk measures and the associated ranges of traffic states; the pNEUMA urban traffic trajectory data[12][13] provide complementary evidence on vehicle kinematics and vehicle-type distributions in urban traffic; and the SPMD connected-vehicle data[14] are used to establish which aspects of the communication process can be observed directly in real-world data and where the corresponding evidential limits lie. The SUMO microscopic traffic-simulation platform[15] is then used to construct the controlled experimental conditions required in this study, including source-risk treatment, opening and closing of the information channel, and physically isolated and physically connected traffic configurations. These real-world datasets are not combined into a single real-world closed-loop causal experiment. Instead, each is used within its own observational scope to provide empirical support for risk measurement, vehicle-motion ranges, communication observability, and the realistic operating range of the simulation, whereas the core counterfactual identification and subsequent control experiments are conducted in the controlled simulation environment. The directly observed variables, specific research roles, and evidential boundaries of each dataset and the simulation platform are summarised in Supplementary Table S1.

The NGSIM data provide the empirical basis for longitudinal car-following risk measurement in this study. We use vehicle-trajectory data from US-101 and Lankershim Boulevard[10][11], from which vehicle position, speed, acceleration, lane information, and the recorded leader relationship are extracted, with all quantities originally reported in imperial units converted to the International System of Units (SI). For each vehicle state with a valid leader record, the corresponding leader is matched at the same time step, and the longitudinal car-following state is reconstructed from inter-vehicle spacing, leader length, and the speeds of the two vehicles. Only states for which the leader can be reliably matched, the net longitudinal gap is positive, and the follower is closing on the leader are included in the car-following conflict-based risk measurement. States for which the leader cannot be matched, the spacing is invalid, or no closing motion is present are retained as unobservable, invalid, or not applicable, respectively, rather than being treated as zero-risk states. The resulting valid car-following states are used to characterise the empirical ranges of longitudinal risk indicators and variables relevant to risk-event identification in real traffic, and to inform the subsequent calibration of risk-measurement rules, event-identification conditions, and the range of traffic states represented in simulation.

The pNEUMA data are used to complement the empirical ranges of vehicle motion and vehicle-type composition under urban traffic conditions. We use the pNEUMA urban traffic trajectory data[12][13], converting the original vehicle-level trajectory records into time-indexed vehicle states and extracting vehicle type, position, speed, longitudinal acceleration, lateral acceleration, and time-related kinematic information. Unlike NGSIM, which is used here to support longitudinal car-following risk measurement, pNEUMA is not used to formally reconstruct leader–follower relationships, calculate TTC or DRAC, or identify car-following risk events. Instead, it is used to characterise the empirical ranges of vehicle speed, acceleration, and vehicle-type distributions in urban traffic. Together with the car-following states derived from NGSIM, these data provide an empirical reference for the plausibility of simulation parameters and traffic states, thereby reducing reliance on a single dataset or a single traffic environment when defining the operating range of the simulation.

The SPMD data are used to define the observability boundaries of communication processes in real-world connected-vehicle data. We use the RV_RX records from SPMD[14], which contain observations of remote vehicle states received by equipped vehicles during operation, including remote-vehicle identifiers, time, position, speed, acceleration, and relative distance. A complete vehicle-device trip is treated as the basic data unit, and the temporal continuity of observations associated with the same remote vehicle is examined to determine which aspects of the receiving side can be supported directly by the available data. Importantly, an individual RV_RX record represents one observation of a remote vehicle state and should not be interpreted as an independent message for which the complete generation, transmission, and reception process is observed. Likewise, a temporal gap between consecutive observations cannot be interpreted directly as packet loss. The available data also contain no validated field indicating whether received information was actually adopted by the vehicle controller. Accordingly, RV_RX is not used here to estimate message-level latency, packet-loss rates, or information adoption, nor to identify the causal effect of remote risk information. Instead, it is

used to establish the observational scope supported by real-world communication data and to distinguish communication properties with direct empirical support from those that must be specified separately as pre-defined conditions in the subsequent simulation experiments.

The SUMO microscopic traffic-simulation platform[15] is used to construct the controlled traffic environment required for the core counterfactual identification and subsequent control experiments. Within each pre-specified experimental scenario, the simulation allows the source-risk treatment and the state of the explicit information channel to be manipulated separately while holding non-treatment factors fixed, and uses scenario design to create either physically isolated or physically connected traffic structures between the source and target regions. Vehicle states, control inputs, message lifecycles, actual vehicle responses, and risk changes are recorded over time, providing the complete experimental records required for Stage I analyses of Effect, Attribution, behavioural response timing, and early risk-outcome timing, as well as a common simulation basis for the information-aware control experiments in Stage II. The ranges of simulated traffic states and vehicle motion are informed by the empirical distributions observed in NGSIM and pNEUMA, whereas SPMD is used to define which aspects of the communication process are supported directly by real-world observations. Communication parameters and information-adoption processes that cannot be identified from the available real-world data are therefore specified and evaluated explicitly as pre-defined simulation conditions rather than interpreted as empirically calibrated quantities.

Risk measurement and source-risk event identification

We define the basic unit of traffic-risk measurement as the relative state of an interacting vehicle pair, rather than the isolated state of a single vehicle considered independently of its traffic environment. For a vehicle v, let v' denote the vehicle with which it forms a direct longitudinal car-following relationship. In the core longitudinal setting considered here, v' is the leader and v is the follower, and $v' \to v$ denotes this ordered leader–follower pair, with v as the vehicle whose risk is being evaluated. This notation specifies the vehicle roles and the direction of risk evaluation; it does not imply that risk is generated unidirectionally by the leader. Instead, the longitudinal collision risk considered here is determined by the relative motion of the two vehicles. Risk quantities are therefore first evaluated at the vehicle-pair level and are subsequently aggregated to the individual-vehicle and target-region levels. It is also important to distinguish a risk value, a risk event, and an actual crash. A risk value characterises the continuously varying level of danger associated with the traffic state, whereas a risk event is a discrete event extracted from a continuous risk trajectory according to pre-specified identification rules, and an actual crash is a possible traffic outcome. Accordingly, throughout this study we maintain the distinction risk value $\neq$ risk event $\neq$ crash and define continuous risk measurement and subsequent source-risk event identification separately.

For an ordered longitudinal leader–follower pair $v' \to v$, we first construct the basic quantities required for risk measurement from the relative motion of the two vehicles. Let $g_t^{(v'\to v)}$ denote the net longitudinal gap between the leader v' and the follower v at time t, defined as the distance from the rear of the leader to the front of the follower along the direction of travel. Let $u_{v,t}$ and $u_{v',t}$ denote the longitudinal speeds of the follower v and leader v', respectively. The closing speed of the follower relative to the leader is defined as

$$\Delta u_t^{(v'\to v)} = u_{v,t} - u_{v',t}. \tag{1}$$

When $\Delta u_t^{(v'\to v)} > 0$, the inter-vehicle gap is closing, that is, the follower is approaching the leader. Such closing motion may result from acceleration of the follower, deceleration of the leader, or concurrent changes in the motion of both vehicles. The risk measure therefore reflects the relative kinematic state of the vehicle pair and does not presume that risk is generated unidirectionally by either vehicle. For states in which the leader can be reliably matched, the net longitudinal gap is positive, and the closing speed is positive, we calculate the time to collision (TTC) and the deceleration rate to avoid a crash (DRAC) as

$$\mathrm{TTC}_t^{(v'\to v)} = \frac{g_t^{(v'\to v)}}{\Delta u_t^{(v'\to v)}}, \tag{2}$$

$$\mathrm{DRAC}_t^{(v'\to v)} = \frac{\left(\Delta u_t^{(v'\to v)}\right)^2}{2g_t^{(v'\to v)}}. \tag{3}$$

A smaller TTC indicates less remaining temporal margin before a potential collision under the current relative motion, whereas a larger DRAC indicates a greater braking demand required to avoid a collision. These indicators enter the subsequent risk calculation only when their physical definition conditions are satisfied. States in which the leader cannot be matched or required inputs are missing are classified as not measurable; states with a non-positive net gap are classified as invalid; and states with a valid leader but no closing motion are classified as not applicable. None of these states is treated as equivalent to zero risk; instead, each is handled separately in the subsequent risk aggregation according to pre-specified validity and missing-data rules.

For vehicle-interaction states that satisfy the validity conditions above, the basic risk indicators, including TTC and DRAC, are transformed according to the mapping ultimately fixed in the Protocol Lock into a directionally consistent instantaneous interaction-risk measure $\rho_t^{(v'\to v)}$, such that larger values always indicate more hazardous traffic states. In the core longitudinal setting, the instantaneous risk of vehicle v at time t, denoted by $\rho_{v,t}$, is determined from its valid longitudinal interaction risk according to the locked rule and is further aggregated over a pre-specified analysis window to obtain the overall vehicle-level risk $R_{v,\omega}$. The aggregate risk of the target region, R_ω^{tgt}, is then obtained by aggregating the vehicle-level risks over the target-vehicle set $\mathcal{V}_\omega^{\mathsf{tgt}}$, whose membership is determined for experimental unit ω according to pre-treatment eligibility rules. The resulting risk-measurement hierarchy is

$$g_t^{(v'\to v)},\ \Delta u_t^{(v'\to v)} \to \mathrm{TTC}_t^{(v'\to v)},\ \mathrm{DRAC}_t^{(v'\to v)} \to \rho_t^{(v'\to v)} \to \rho_{v,t} \to R_{v,\omega} \to R_\omega^{\mathsf{tgt}}.$$

To ensure that risk differences are comparable across experimental conditions, target-vehicle eligibility, analysis win-

dows, the handling of valid, not-applicable, invalid, and not-measurable states, and the aggregation rules at both the vehicle and regional levels are all specified in advance and applied consistently. In particular, target vehicles are not reselected on the basis of whether they subsequently receive or adopt remote information, exhibit a vehicle response, or experience a change in risk.

A continuous risk state does not, by itself, constitute a distinct risk event. For experimental unit ω, the specified source-risk event that ultimately triggers the remote information pathway is denoted by $\eta_{\omega}^{\mathsf{src}}$. This event is extracted from the continuous risk trajectory of an ordered leader–follower pair $v' \to v$ within the source region according to pre-specified event-identification rules, rather than being defined by a high risk value observed at a single time step. The event-identification rules must specify, at a minimum, the triggering indicator and its threshold, the minimum event duration, event-entry and event-exit conditions, and the rules for merging and deduplicating contiguous or closely spaced risk episodes. When the same vehicle pair satisfies the risk condition over multiple consecutive or sufficiently closely spaced time steps, whether these observations constitute a single risk event is determined by the pre-specified continuity, merging, and deduplication rules; individual time steps satisfying the condition are not automatically counted as separate events. If multiple candidate risk events occur within the same experimental unit, a pre-specified selection rule is further used to determine which event is designated as the source-risk event for the subsequent information-propagation experiment.

The experimental source-risk treatment must be distinguished from the actual formation of a source-risk event. For experimental unit ω, let $t_{\omega}^{\mathsf{src\text{-}int}}$ denote the pre-specified time at which the source-risk treatment begins, and let $t_{\omega}^{\mathsf{evt}}$ denote the time at which the specified source-risk event $\eta_{\omega}^{\mathsf{src}}$ is first formally established according to the locked event-identification rules. These times generally satisfy

$$t_{\omega}^{\mathsf{src\text{-}int}} \le t_{\omega}^{\mathsf{evt}}. \tag{4}$$

The former marks the point at which the experiment begins to alter traffic conditions in the source region, whereas the latter marks the point at which the observed risk trajectory first satisfies the pre-specified criteria for a qualifying source-risk event. Applying the source-risk treatment therefore does not guarantee that $\eta_{\omega}^{\mathsf{src}}$ will occur. If the treatment has been applied but no qualifying event is formed within the prescribed window, the experimental unit is handled according to the pre-specified quality-control and inclusion or exclusion rules; the event threshold or identification criteria are not relaxed after observing subsequent outcomes in the target region. Only after $\eta_{\omega}^{\mathsf{src}}$ has been formally established at $t_{\omega}^{\mathsf{evt}}$ does it become eligible to trigger the subsequent generation of remote risk information.

The risk-measurement, aggregation, and event-identification rules described above are calibrated, validated, and incorporated into the Protocol Lock before Stage I begins. The items to be specified in advance include the final definition and parameters of the instantaneous risk measure; the criteria and handling rules for valid, not-applicable, invalid, and not-measurable states; the analysis windows and eligibility criteria for evaluable states; the pre-treatment eligibility rules for target vehicles; the aggregation procedures for vehicle-level and target-region risk; and the triggering indicator, threshold, duration requirement, entry and exit conditions, merging and deduplication rules, and multiple-candidate selection procedure for source-risk events. The corresponding quality-control criteria and rules for retaining or excluding experimental units are specified at the same stage. Once the Protocol Lock is completed, these rules are applied consistently across all Stage I counterfactual conditions and are not modified in response to the magnitude, statistical significance, or direction of the observed target-region effects. This ensures that risk differences observed in subsequent experiments are evaluated under a common risk definition, analysis population, and event-identification protocol, rather than arising from post hoc changes to measurement or selection criteria.

Physical propagation, message lifecycle and temporal ordering

For a specified source-risk event $\eta_{\omega}^{\mathsf{src}}$ established within experimental unit ω, its subsequent influence on the target region may occur through two distinct pathways. The first is ordinary physical propagation through car-following, merging, and other local traffic interactions; the second is an explicit information pathway through which risk information tied to the same source event is transmitted to target vehicles and, once adopted, enters their decision-making processes. To compare when these two pathways first produce actual behavioural effects in the target region, we separately define the earliest time at which ordinary physical propagation induces an attributable actual vehicle response in the target region and the earliest time at which remote information induces such a response. Both responses must be traceable to the same specified source-risk event, ensuring that the temporal comparison concerns two competing pathways originating from a common source rather than unrelated traffic changes. The two propagation processes and their corresponding timing rules are defined below.

To represent how ordinary traffic effects may propagate progressively through local vehicle interactions, we describe the physical interaction structure at time t as a dynamic directed graph

$$G^{\mathsf{phy}}(t) = \left(\mathcal{V}, E^{\mathsf{phy}}(t)\right), \tag{5}$$

where $(v_i \to v_j) \in E^{\mathsf{phy}}(t)$ indicates that, under the current traffic state, vehicle v_i is capable of directly influencing vehicle v_j through an ordinary local traffic interaction, such as car-following, merging, or conflict interaction. A physical edge represents a potential direct interaction and does not by itself imply that a disturbance originating from the specified source-risk event has actually propagated along that edge. For an actual single-hop physical propagation to be established, vehicle v_i must first exhibit a state change attributable to the relevant disturbance, after which vehicle v_j must show an attributable response through the corresponding local interaction, with the required temporal ordering preserved. In a longitudinal car-following setting, this typically means that the leader first exhibits a relevant change in speed, acceleration, or another traffic state, followed by an actual response of the follower through the car-following interaction. Multi-hop physical propagation is defined as a temporally ordered sequence of such valid single-hop propagations, with each consecutive vehicle pair satisfying both the actual-interaction

requirement and a consistent direction of time. Road connectivity or vehicle adjacency at a single time point alone is therefore insufficient to establish that physical propagation has occurred.

On this basis, for any ordinary physical-propagation process originating from the specified source-risk event and ultimately producing an actual vehicle response in the target region, we retain only the portion of the propagation process from the source up to the first attributable physical response within the target region and represent it as a temporally ordered multi-hop propagation chain

$$\mathcal{P}^{\mathsf{phy}} = (v_0, v_1, \ldots, v_k). \tag{6}$$

Each hop between consecutive vehicles must satisfy the actual physical-propagation conditions defined above: a relevant state change in the preceding vehicle must induce a subsequent attributable response in the following vehicle through the corresponding local traffic interaction. Vehicle v_k is the first vehicle along the chain that lies within the target region and exhibits an attributable response to the source-originating physical propagation, whereas $v_0, \ldots, v_{k-1}$ remain outside the target region. Thus, $v_{k-1} \rightarrow v_k$ represents the physical interaction through which the propagation chain first produces an actual vehicle response within the target region. Let

$$t^{\mathsf{resp}}\left(\mathcal{P}^{\mathsf{phy}}\right) \tag{7}$$

denote the time at which vehicle v_k first exhibits an attributable actual response to the physical influence transmitted from v_{k-1}. Let $\mathfrak{P}_{\omega}^{\mathsf{phy}}$ denote the set of all valid physical-propagation chains satisfying these conditions in experimental unit ω. The earliest physical-response time is then defined as

$$t_{\omega}^{\mathsf{phy\text{-}resp}} = \min_{\mathcal{P}^{\mathsf{phy}} \in \mathfrak{P}_{\omega}^{\mathsf{phy}}} t^{\mathsf{resp}}\left(\mathcal{P}^{\mathsf{phy}}\right). \tag{8}$$

This quantity identifies the earliest time at which ordinary physical propagation attributable to the specified source-risk event produces an actual vehicle response anywhere within the target region. The minimum is taken across all valid source-to-target physical-propagation chains because the temporal comparison concerns the earliest point at which the physical pathway can exert a behavioural influence on the target region; selecting a slower pathway would provide an inappropriate reference for comparison with the information pathway. In a physically isolated setting, if no valid ordinary physical-propagation chain exists from the source region to the target region, such that $\mathfrak{P}_{\omega}^{\mathsf{phy}} = \varnothing$, we define

$$t_{\omega}^{\mathsf{phy\text{-}resp}} = +\infty. \tag{9}$$

Physical isolation here refers specifically to the ordinary physical-propagation pathway from the source region to the target region and does not preclude normal car-following or other local traffic interactions among vehicles within the target region itself.

For the information pathway, we represent the explicit communication structure at time t as a dynamic directed graph

$$G^{\mathsf{info}}(t) = \left(\mathcal{V}, E^{\mathsf{info}}(t)\right), \tag{10}$$

where $(v_i \rightarrow v_j) \in E^{\mathsf{info}}(t)$ indicates that, under the current communication configuration and experimental protocol, information associated with vehicle v_i is permitted to be transmitted explicitly to vehicle v_j at time t. This relation describes communication reachability or protocol-level permission rather than a local physical traffic interaction, and therefore, in general,

$$G^{\mathsf{phy}}(t) \neq G^{\mathsf{info}}(t). \tag{11}$$

The existence of an information edge does not, by itself, imply that an actual information-mediated influence has occurred. Whether a specific message is generated and subsequently enters the vehicle's decision-making and behavioural processes is determined separately from the recorded message lifecycle and vehicle-response evidence defined below.

Building on this communication structure, remote risk information must also have a clearly defined and traceable source-event origin. A risk message q tied to the specified source-risk event $\eta_{\omega}^{\mathsf{src}}$ may be generated only after that event has been formally established according to the preceding rules and in accordance with the pre-specified message-generation protocol. An open information channel therefore does not imply that a message has actually been generated. For message q, we distinguish four stages of its subsequent lifecycle: generation, transmission, delivery, and adoption, with the corresponding times denoted by

$$t_q^{\mathsf{gen}}, \qquad t_q^{\mathsf{send}}, \qquad t_{q,v}^{\mathsf{del}}, \qquad t_{q,v}^{\mathsf{adopt}},$$

representing, respectively, the completion of message generation, its actual entry into the communication channel, its delivery to target vehicle v, and the first time at which it is accepted by the decision or control module of vehicle v as a valid input. For a message that is ultimately adopted by vehicle v, the lifecycle must satisfy

$$t_{\omega}^{\mathsf{evt}} \leq t_q^{\mathsf{gen}} \leq t_q^{\mathsf{send}} \leq t_{q,v}^{\mathsf{del}} \leq t_{q,v}^{\mathsf{adopt}}. \tag{12}$$

These stages are conceptually distinct: message generation does not imply transmission, transmission does not imply delivery, and delivery does not imply adoption. The lifecycle record therefore retains not only complete successful pathways but also interrupted states, including messages that are generated but not transmitted, transmitted but not delivered, rendered invalid before use, or delivered but not adopted. This prevents the structural availability of a communication channel or a partially completed message-transfer process from being mistaken for an actual information-mediated influence. Except for communication conditions explicitly manipulated as experimental factors, the message-generation rules, lifecycle-state definitions, adoption criteria, and corresponding logging procedures are all specified before Stage I and applied consistently across comparable experimental conditions.

Message adoption does not necessarily imply that an actual behavioural response has occurred. For a target vehicle v, once message q is adopted at $t_{q,v}^{\mathsf{adopt}}$, it first enters the vehicle's decision-making or control process and may alter the requested control input $x_{v,t}^{\mathsf{cmd}}$. This request is then processed through the pre-specified safety-handling and actuation mechanisms to produce the actually executed control input $x_{v,t}^{\mathsf{act}}$, which subsequently affects the vehicle's physical motion through the vehicle dynamics and may alter its later risk state. We therefore distinguish information adoption, changes in requested control, actually executed control,

and the resulting vehicle response. Let $t_{q,v}^{\mathsf{resp}}$ denote the earliest time after adoption at which vehicle v exhibits an actual control response attributable to message q. Accordingly,

$$t_{q,v}^{\mathsf{adopt}} \le t_{q,v}^{\mathsf{resp}}. \tag{13}$$

An actual response is not defined as an arbitrary change in speed or acceleration observed in the vehicle trajectory, because such changes may also arise in the absence of remote information through car-following interactions, road conditions, or other local traffic influences. A response is classified as information-induced only when, according to the pre-specified response-identification rules, the executed control or the resulting change in vehicle motion can be linked to the control difference arising from adoption of the message. This classification does not depend on whether target-region risk subsequently changes. The pathway from remote information entering the vehicle's decision process to its effect on the physical traffic state can therefore be represented as

$$q \rightarrow \text{adoption} \rightarrow x_{v,t}^{\mathsf{cmd}} \rightarrow x_{v,t}^{\mathsf{act}} \rightarrow \text{vehicle motion state} \rightarrow \rho_{v,t}.$$

For the specified source-risk event $\eta_\omega^{\mathsf{src}}$, we next determine the earliest actual information-induced response across all messages tied to that event and all target vehicles pre-specified as eligible to receive and adopt such messages. Let $\mathcal{V}_\omega^{\mathsf{msg}}$ denote the set of message-eligible target vehicles in experimental unit ω, with $\mathcal{V}_\omega^{\mathsf{msg}} \subseteq \mathcal{V}_\omega^{\mathsf{tgt}}$. Let $I_{\mathcal{R}^{\mathsf{tgt}}}(v,t)$ be an indicator of target-region membership, taking the value 1 when vehicle v is located within $\mathcal{R}^{\mathsf{tgt}}$ at time t and 0 otherwise. The earliest information-response time is then defined as

$$t_\omega^{\mathsf{info\text{-}resp}} = \min_{\substack{q \leftrightarrow \eta_\omega^{\mathsf{src}} \\ v \in \mathcal{V}_\omega^{\mathsf{msg}} \\ I_{\mathcal{R}^{\mathsf{tgt}}}\left(v, t_{q,v}^{\mathsf{resp}}\right)=1}} t_{q,v}^{\mathsf{resp}}. \tag{14}$$

Here, $q \leftrightarrow \eta_\omega^{\mathsf{src}}$ indicates that message q can be uniquely traced to and associated with the specified source-risk event. The quantity $t_\omega^{\mathsf{info\text{-}resp}}$ therefore represents the earliest time at which remote information generated from that source-risk event produces an attributable actual response in an eligible vehicle while that vehicle is within the target region. We use $\mathcal{V}_\omega^{\mathsf{msg}}$ rather than the full target-vehicle set $\mathcal{V}_\omega^{\mathsf{tgt}}$ because this timing quantity is intended to capture the first point at which the remote information pathway directly enters the behavioural layer of the target region. Target vehicles that are not message-eligible do not contribute to this timing definition, although their subsequent risk may still contribute to aggregate target-region risk and may be influenced indirectly through normal physical interactions within the target region. If no qualifying attributable response is observed within the pre-specified observation window, the corresponding information-response time is retained as unobserved or right-censored and handled according to the pre-specified statistical procedure, rather than being assigned an arbitrary value such as $+\infty$.

After determining the earliest information-response time $t_\omega^{\mathsf{info\text{-}resp}}$ and the earliest physical-response time $t_\omega^{\mathsf{phy\text{-}resp}}$, we compare the temporal ordering of the two pathways at the level of actual vehicle response in the target region. Experimental unit ω is classified as satisfying the information-first criterion if and only if

$$t_\omega^{\mathsf{info\text{-}resp}} < t_\omega^{\mathsf{phy\text{-}resp}}. \tag{15}$$

A strict inequality is used because, when the two response times are equal, the available temporal resolution does not establish that the response attributable to the information pathway occurred before that attributable to ordinary physical propagation. Such cases are therefore not classified as information-first.

Under physically connected conditions, when both response times are finite, we further define the information lead time as

$$\Delta t_\omega^{\mathsf{lead}} = t_\omega^{\mathsf{phy\text{-}resp}} - t_\omega^{\mathsf{info\text{-}resp}}. \tag{16}$$

A positive value, $\Delta t_\omega^{\mathsf{lead}} > 0$, indicates that the specified remote information produces an actual vehicle response in the target region earlier than ordinary physical propagation. A value of $\Delta t_\omega^{\mathsf{lead}} = 0$ indicates that the available temporal resolution does not establish an ordering between the two responses, whereas $\Delta t_\omega^{\mathsf{lead}} < 0$ indicates that ordinary physical propagation produces the earlier target-region response.

Under physically isolated conditions, the ordinary physical-propagation pathway from the source region to the target region is structurally removed, so that $t_\omega^{\mathsf{phy\text{-}resp}} = +\infty$. Accordingly, any finite $t_\omega^{\mathsf{info\text{-}resp}}$ satisfies the information-first criterion defined above. The primary purpose of the physically isolated setting, however, is not to quantify a finite information lead time, but to exclude ordinary source-to-target physical propagation as a competing explanation for the observed remote information effect. Quantitative comparison of finite lead times is therefore primarily conducted under physically connected conditions, where both the information and physical pathways are able to produce actual responses in the target region.

It is important to emphasise that the information-first criterion establishes only the temporal ordering of the two pathways at the behavioural-response level and therefore corresponds to the behavioural component of Temporal precedence in the evidential framework. By itself, it does not establish that aggregate risk in the target region has changed, nor does it demonstrate that any regional risk difference emerged before ordinary physical propagation began to influence the corresponding risk outcome. The latter constitutes a separate risk-outcome-level timing requirement and is evaluated using an independently defined temporal boundary and risk-evaluation rule. Accordingly, $t_\omega^{\mathsf{phy\text{-}resp}}$ is used only to compare the earliest attributable behavioural responses of the physical and information pathways and is not used directly as the boundary of the early risk-evaluation window.

Stage I: Remote risk-information effects and attribution under physically isolated conditions

Stage I first tests whether specified remote risk information produces an independent and attributable incremental effect on the aggregate risk of the target region. To exclude ordinary physical propagation from the source region to the target region as far as possible as a competing explanation, the core counterfactual identification is first conducted under physically isolated conditions. In this setting, the ordinary physical-propagation pathway from the source region to the target region is structurally removed, while normal vehicle interactions within the target region are retained. Consequently, differences in target-region risk cannot be directly explained by ordinary physical propagation from the source,

providing a clearer experimental setting in which to identify the regional-level Effect of remote risk information and establish its Attribution. We next define the counterfactual treatment factors, the experimental unit, and the core effect measures used for this identification.

Stage I includes two binary experimental factors that can be manipulated separately. The first is the source-risk treatment state, denoted by

$$\mathsf{scr} \in \{0, 1\}, \tag{17}$$

where $\mathsf{scr} = 0$ indicates that the pre-specified source-risk treatment is not applied and $\mathsf{scr} = 1$ indicates that it is applied. Importantly, $\mathsf{scr} = 1$ denotes only the experimental application of the source-risk treatment and does not imply that the specified source-risk event $\eta_\omega^{\mathsf{src}}$ necessarily occurs; whether such an event is actually established is determined from the realised risk trajectory using the locked event-identification rules defined above. The second factor is the state of the explicit remote-information channel, denoted by

$$\mathsf{opn} \in \{0, 1\}, \tag{18}$$

where $\mathsf{opn} = 0$ indicates that the remote-information channel is closed and $\mathsf{opn} = 1$ indicates that it is open. An open channel means only that the experimental protocol permits information associated with a specified source-risk event to be transmitted; it does not imply that a message has been generated, transmitted, delivered, adopted, or has produced an actual vehicle response. The two factors define four experimental conditions,

$$(\mathsf{scr}, \mathsf{opn}) \in \{0, 1\} \times \{0, 1\}, \tag{19}$$

namely $(0, 0)$, $(0, 1)$, $(1, 0)$, and $(1, 1)$. Condition $(1, 1)$ applies the source-risk treatment and opens the remote-information channel; $(1, 0)$ retains the source-risk treatment while closing the channel; $(0, 1)$ opens the channel without applying the source-risk treatment; and $(0, 0)$ applies neither factor. This 2×2 factorial design separates the source-risk treatment from information-channel opening and thereby allows the subsequent analysis to determine whether the effect associated with opening the information channel changes systematically with the source-risk treatment state (Fig. 2a).

Under these four experimental conditions, Stage I treats a complete set of four matched runs as the basic experimental unit, denoted by $\omega \in \Omega$. Each experimental unit ω comprises the $(0, 0)$, $(0, 1)$, $(1, 0)$, and $(1, 1)$ conditions executed under the same pre-specified traffic scenario and matched stochastic conditions, rather than representing any single simulation run. Each of the four runs is independently initialised and does not share vehicle states, message states, or other dynamic simulation states with the others during execution. Apart from the two manipulated factors scr and opn, the road and regional definitions, initial traffic conditions, baseline vehicle models and control rules, risk measurement and target-vehicle eligibility criteria, analysis-time rules, and all non-treatment components of the communication and control protocols are held fixed according to the pre-specified design. For stochastic traffic behaviour, the four runs within the same ω use the same master random seed, with corresponding stochastic processes paired and aligned as far as practicable to minimise background variation in traffic and driving behaviour. The information-channel state must not feed back to alter the source-risk treatment or the mechanism by which the source-risk event is generated. Consequently, the specified source-risk events arising from the same source-risk treatment in conditions $(1, 0)$ and $(1, 1)$ must satisfy the pre-specified matching rules for both event identity and establishment time; otherwise, any difference in target-region outcomes between the two conditions could also reflect differences in the source event and could not be attributed to the information-channel state alone. These consistency requirements apply to pre-treatment conditions and non-treatment factors only. Divergence in vehicle trajectories, message states, and risk outcomes after the manipulated conditions take effect is part of the experimental response and is not required to remain matched.

The core four-condition effect is estimated only from complete experimental units that satisfy the pre-specified validity requirements. Let Ω^{valid} denote the set of experimental units meeting these requirements. Inclusion in this set requires, at a minimum, that condition $(1, 0)$ produces a specified source-risk event satisfying the locked event-identification rules, that condition $(1, 1)$ contains a corresponding source-risk event satisfying the pre-specified matching criteria, and that all four experimental conditions pass the pre-specified quality-control and evaluability checks. If any condition exhibits a serious quality-control failure that renders the core outcome unevaluable, or if the two source-risk-treatment conditions fail to produce a valid matched pair of source-risk events, the complete experimental unit is excluded from estimation of the core four-condition effect; the remaining conditions are not retained and recombined into a partial comparison. All rules governing event matching, quality control, and the inclusion or exclusion of experimental units are fixed before Stage I begins.

To ensure that target-region risk is compared across the four conditions at the same event-relative time position and over fixed observation windows, Stage I establishes a common temporal reference for each complete experimental unit (Fig. 2b). For each $\omega \in \Omega^{\mathsf{valid}}$, the establishment time t_ω^{evt} of the specified source-risk event $\eta_\omega^{\mathsf{src}}$ is determined under condition $(1, 0)$, in which the source-risk treatment is applied while the remote-information channel is closed, using the locked event-identification rules defined above. This time is then used as the common temporal reference for risk analysis across all four conditions. Condition $(1, 0)$ is used to define this reference because it preserves the source-risk treatment while excluding any target-side influence arising through the remote-information channel, thereby providing a source-event time that is not affected by feedback from the information pathway. Condition $(1, 1)$ does not define a separate analysis time; instead, its corresponding source-risk event must satisfy the pre-specified matching rules for event identity and establishment time relative to the event identified in $(1, 0)$. For conditions $(0, 0)$ and $(0, 1)$, in which the source-risk treatment is absent, the same clock position from the matched experimental unit is used solely for comparison and is not interpreted as indicating that a source-risk event actually occurred in those conditions.

Using this common temporal reference, we first define a reference interval during which remote information triggered by the specified source-risk event cannot yet have produced an actual effect. The pre-information reference window is

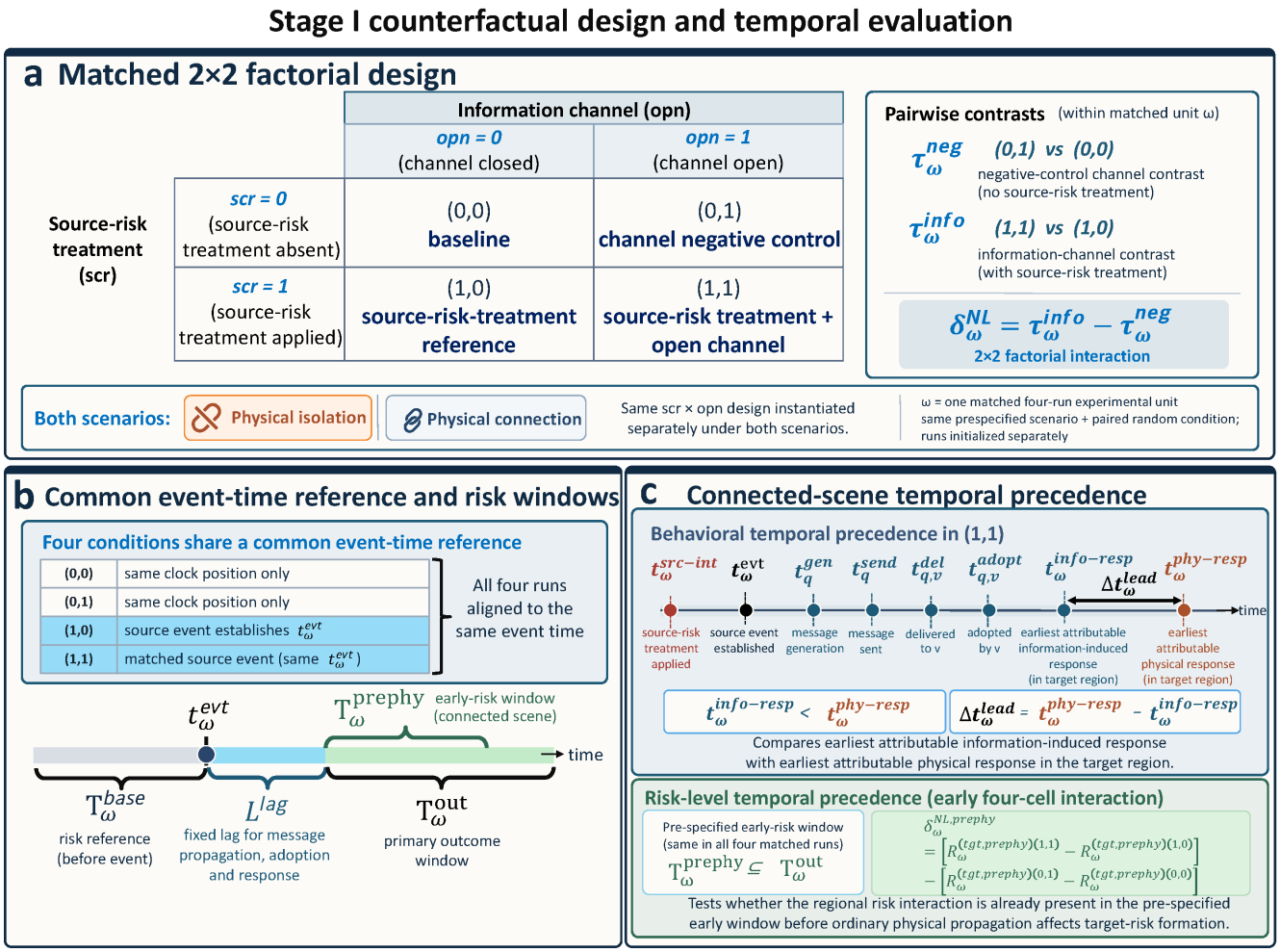


Figure 2: Stage I counterfactual identification and temporal-evaluation design. **a,** The source-risk treatment state scr and the explicit information-channel state opn define a matched 2×2 factorial design; the same four-condition design is applied under both physically isolated and physically connected settings, and the four matched runs together constitute experimental unit ω. **b,** The four conditions are aligned by the common source-event time t_ω^{evt} determined from condition $(1,0)$, and share a common pre-information reference window, a fixed waiting period before information influence, and a core outcome window; under physically connected conditions, a pre-specified early risk-evaluation window $\mathcal{T}_\omega^{\text{prephy}}$ is further defined at the beginning of the core outcome window. **c,** Under the physically connected $(1,1)$ condition, the earliest actual target-region vehicle responses induced by the information pathway and by ordinary physical propagation are recorded separately for the same specified source-risk event, and behavioural-level information-first is determined by $t_\omega^{\text{info-resp}} < t_\omega^{\text{phy-resp}}$. In parallel, the early four-condition risk interaction effect is re-estimated within the shared $\mathcal{T}_\omega^{\text{prephy}}$ to test whether the target-region risk effect has already emerged before ordinary physical propagation from the source begins to influence risk formation.

defined as

$$\mathcal{T}_\omega^{\text{base}} = \left[t_\omega^{\text{evt}} - L^{\text{base}},\ t_\omega^{\text{evt}}\right), \tag{20}$$

where L^{base} is the pre-specified length of the reference window. Because information tied to the source-risk event can be generated only after the event has been formally established and must then proceed through message generation, transmission, delivery, adoption, and an actual vehicle response, remote information should not be assumed to alter target-region risk immediately after t_ω^{evt}. We therefore introduce a fixed waiting period L^{lag} after the source-risk event to allow time for message propagation, information adoption, and the resulting vehicle response, and define the core outcome window after this waiting period as

$$\mathcal{T}_\omega^{\text{out}} = \left[t_\omega^{\text{evt}} + L^{\text{lag}},\ t_\omega^{\text{evt}} + L^{\text{lag}} + L^{\text{out}}\right), \tag{21}$$

where L^{out} is the pre-specified length of the core outcome window.

For the physically connected setting, we further pre-specify an early risk-evaluation window at the beginning of the core outcome window,

$$\mathcal{T}_\omega^{\text{prephy}} = \left[t_\omega^{\text{evt}} + L^{\text{lag}},\ t_\omega^{\text{evt}} + L^{\text{lag}} + L^{\text{prephy}}\right), \tag{22}$$

where L^{prephy} is the pre-specified duration of the early risk-evaluation window and satisfies

$$0 < L^{\text{prephy}} \le L^{\text{out}}, \tag{23}$$

such that

$$\mathcal{T}_\omega^{\text{prephy}} \subseteq \mathcal{T}_\omega^{\text{out}}. \tag{24}$$

The early risk-evaluation window $\mathcal{T}_\omega^{\text{prephy}}$ is used to test whether the target-region risk effect has already emerged before ordinary physical propagation originating from the specified source-risk event begins to influence risk formation in

the target region. Its temporal position and duration are determined before Stage I on the basis of the pre-specified spatial configuration of the experimental scenario, constraints on ordinary physical propagation, and independent calibration results, and the same early evaluation window is applied to all four conditions within experimental unit ω. The quantities L^{base}, L^{lag}, L^{out}, and L^{prephy} are all incorporated into the Protocol Lock before Stage I begins. They are not shifted or shortened post hoc on the basis of message delivery, information adoption, information-induced responses, physical responses, or changes in risk observed during any of the four experimental runs.

Under this common temporal structure, target-region risk is calculated for all four conditions using the locked risk-measurement, target-vehicle eligibility, and regional-aggregation rules defined above. For experimental condition $(\mathsf{scr}, \mathsf{opn}) = (s, o)$, where $s, o \in \{0, 1\}$, let

$$R_{\omega}^{\mathsf{tgt},\mathsf{base}}(s, o) \tag{25}$$

denote the aggregate target-region risk evaluated over the pre-information reference window $\mathcal{T}_{\omega}^{\mathsf{base}}$, and let

$$R_{\omega}^{\mathsf{tgt},\mathsf{out}}(s, o) \tag{26}$$

denote the aggregate target-region risk evaluated over the core outcome window $\mathcal{T}_{\omega}^{\mathsf{out}}$ following the fixed waiting period. The former characterises the reference level of target-region risk before remote information associated with the specified source-risk event could plausibly have produced an actual effect, whereas the latter characterises target-region risk after the source-risk event has been established and a fixed interval has been allowed for message propagation, information adoption, and subsequent vehicle responses.

For the physically connected setting, target-region risk is also evaluated over the pre-specified early risk-evaluation window $\mathcal{T}_{\omega}^{\mathsf{prephy}}$ using exactly the same instantaneous risk definition, target-vehicle eligibility criteria, validity handling, and regional-aggregation rules as those applied to the core outcome window. We denote this quantity by

$$R_{\omega}^{\mathsf{tgt},\mathsf{prephy}}(s, o). \tag{27}$$

Accordingly, $R_{\omega}^{\mathsf{tgt},\mathsf{prephy}}(s, o)$ and $R_{\omega}^{\mathsf{tgt},\mathsf{out}}(s, o)$ differ only in the pre-specified temporal range over which target-region risk is evaluated; the risk measure, analysis population, and four-condition counterfactual structure remain unchanged.

We further define, for each individual experimental condition, the change in target-region risk relative to its pre-information reference level as

$$\Delta R_{\omega}^{\mathsf{tgt}}(s, o) = R_{\omega}^{\mathsf{tgt},\mathsf{out}}(s, o) - R_{\omega}^{\mathsf{tgt},\mathsf{base}}(s, o). \tag{28}$$

The quantity $\Delta R_{\omega}^{\mathsf{tgt}}(s, o)$ is used to examine how risk changes relative to the reference level within each condition and serves as an auxiliary outcome for baseline adjustment and robustness analyses. The primary outcome for the core Stage I four-condition counterfactual identification nevertheless remains $R_{\omega}^{\mathsf{tgt},\mathsf{out}}(s, o)$, evaluated over the common core outcome window. For notational simplicity, unless otherwise specified, we therefore write

$$R_{\omega}^{\mathsf{tgt}}(s, o) \equiv R_{\omega}^{\mathsf{tgt},\mathsf{out}}(s, o). \tag{29}$$

The early-window quantity $R_{\omega}^{\mathsf{tgt},\mathsf{prephy}}(s, o)$ is used only for the risk-outcome-level temporal-precedence test under physically connected conditions and does not replace the primary outcome.

Within the common core outcome window, we first examine the target-region risk difference associated with opening the remote-information channel separately in the presence and absence of the source-risk treatment. When the source-risk treatment is applied, we define the source-risk-present channel contrast for experimental unit ω as

$$\tau_{\omega}^{\mathsf{info}} = R_{\omega}^{\mathsf{tgt}}(1, 1) - R_{\omega}^{\mathsf{tgt}}(1, 0). \tag{30}$$

This contrast compares two conditions with the same source-risk treatment that differ only in the state of the information channel, and therefore characterises the difference in target-region risk associated with opening the remote-information channel when the source-risk treatment is present. Because $\mathsf{opn} = 1$ still indicates only that the channel is permitted to operate, this contrast alone does not establish that the observed risk difference is mediated by the actual adoption of the specified message and the resulting vehicle response. Under the common risk-direction convention used in this study, $\tau_{\omega}^{\mathsf{info}} < 0$ indicates lower target-region risk when the information channel is open, whereas $\tau_{\omega}^{\mathsf{info}} > 0$ indicates higher target-region risk.

At the same time, when the source-risk treatment is absent, we define the negative-control channel contrast as

$$\tau_{\omega}^{\mathsf{neg}} = R_{\omega}^{\mathsf{tgt}}(0, 1) - R_{\omega}^{\mathsf{tgt}}(0, 0). \tag{31}$$

When $\mathsf{scr} = 0$, no risk message is triggered by the specified source-risk event. This contrast is therefore not used to estimate the effect of the specified remote risk information itself, but rather to test whether merely opening the information channel, or other implementation differences associated with channel state, systematically alters target-region risk. If changing the channel state introduces no systematic difference other than permitting remote-information transmission, then $\tau_{\omega}^{\mathsf{neg}}$ is expected to be close to zero, although this is not guaranteed by the experimental definition. A substantial deviation from zero would therefore motivate further examination of the channel implementation, baseline control logic, stochastic processes, or other unintended influences. Accordingly, $\tau_{\omega}^{\mathsf{info}}$ and $\tau_{\omega}^{\mathsf{neg}}$ represent the channel contrasts in the presence and absence of the source-risk treatment, respectively, with the latter serving as a negative control for channel opening in the subsequent core four-condition interaction analysis.

The source-risk-present channel contrast $\tau_{\omega}^{\mathsf{info}}$ alone cannot distinguish a channel-related difference that is specific to the presence of the source-risk treatment from a background difference that may arise simply from opening the information channel. We therefore compare the channel contrasts obtained with and without the source-risk treatment. Let the superscript NL denote the nonlocal information effect considered in this study. The core four-condition interaction effect for experimental unit ω is defined as

$$\delta_{\omega}^{\mathsf{NL}} = \tau_{\omega}^{\mathsf{info}} - \tau_{\omega}^{\mathsf{neg}}, \tag{32}$$

that is,

$$\delta_{\omega}^{\mathsf{NL}} = \left[R_{\omega}^{\mathsf{tgt}}(1, 1) - R_{\omega}^{\mathsf{tgt}}(1, 0)\right] - \left[R_{\omega}^{\mathsf{tgt}}(0, 1) - R_{\omega}^{\mathsf{tgt}}(0, 0)\right]. \tag{33}$$

This quantity measures the additional change in the information-channel contrast when the source-risk treatment is present relative to the negative-control channel contrast observed when the source-risk treatment is absent. It is therefore a 2×2 factorial interaction contrast between source-risk treatment and information-channel opening, rather than a classical difference-in-differences estimator based on treatment-group and before–after comparisons. Because larger risk values indicate more hazardous traffic states, $\delta_{\omega}^{\mathsf{NL}} < 0$ indicates an additional reduction in target-region risk associated with opening the remote-information channel under the source-risk treatment, whereas $\delta_{\omega}^{\mathsf{NL}} > 0$ indicates an additional increase in risk; no effect direction is assumed a priori. Across the pre-specified population of complete valid experimental units, the overall four-condition interaction effect is defined as

$$\Delta^{\mathsf{NL}} = \mathbb{E}\left[\delta_{\omega}^{\mathsf{NL}} \mid \omega \in \Omega^{\mathsf{valid}}\right]. \tag{34}$$

Thus, $\delta_{\omega}^{\mathsf{NL}}$ and Δ^{NL} represent the experimental-unit-level and population-level regional risk interaction effects, respectively. They characterise the incremental remote-information effect specifically associated with the source-risk treatment, rather than the risk change in any single run, individual vehicle, or isolated time point. Importantly, this four-condition interaction provides only one component of the Effect evidence and does not by itself establish the full phenomenon of information-mediated action at a distance. Attribution to the specified information and Temporal precedence relative to ordinary physical propagation require separate supporting evidence as defined above. Finite-sample estimation, statistical uncertainty, between-unit heterogeneity, and criteria for practical significance are specified later in the statistical analysis.

The four-condition interaction effect defined above provides regional-level Effect evidence, but a non-zero $\delta_{\omega}^{\mathsf{NL}}$ or overall interaction effect Δ^{NL} does not by itself establish Attribution to the specified remote risk information. Attribution additionally requires that the observed difference in target-region risk can be traced through the end-to-end information-action pathway to the same specified source-risk event: the corresponding message must actually be generated, transmitted, delivered, and adopted, enter the target vehicle's decision-making and control process, produce an attributable actual vehicle response, and maintain a mechanistically consistent relationship with the subsequent change in risk. The negative-control contrast $(0,1) - (0,0)$ is used in parallel to identify potential spurious effects arising from merely opening the information channel or from other implementation differences associated with channel state. Stage I Attribution is therefore supported jointly by the regional-level four-condition counterfactual evidence and the traceable information-action pathway defined above. The former establishes whether the experimental factors produce an incremental difference in target-region risk, whereas the latter establishes whether that difference is mechanistically consistent with the specified remote information; neither form of evidence can substitute for the other.

Taken together, the physically isolated setting establishes the core counterfactual framework for identifying Effect and Attribution in Stage I. Effect is characterised by the complete four-condition interaction contrast, whereas Attribution is supported jointly by the counterfactual comparison and the traceable information-action pathway. Because this setting structurally removes ordinary physical propagation from the source region to the target region, however, it cannot test how the information-mediated effect behaves when ordinary physical propagation is simultaneously present as a competing pathway. The next subsection therefore restores the ordinary physical-propagation pathway and, while retaining the same definitions of Effect and Attribution, extends the analysis to a joint test under physically connected conditions.

Stage I: Remote risk-information effects and response timing under physically connected conditions

The physically isolated setting provides the core identification of the effect of specified remote risk information and its Attribution while excluding ordinary physical propagation from the source region to the target region as far as possible as a competing explanation. However, because ordinary physical propagation from the source cannot produce an actual vehicle response in the target region under this setting, it cannot determine whether the identified information effect persists when the information and physical pathways coexist, nor can it establish their temporal ordering when both are capable of influencing the target region. Stage I therefore further restores the ordinary physical-propagation pathway from the source region to the target region. Under these physically connected conditions, we test whether the previously identified remote information effect remains detectable in the presence of competing physical propagation and whether the information pathway produces an actual vehicle response in the target region before ordinary physical propagation from the same source-risk event does so.

The defining structural change in the physically connected setting is the restoration of ordinary physical reachability from the source region $\mathcal{R}^{\mathsf{src}}$ to the target region $\mathcal{R}^{\mathsf{tgt}}$, while the definitions of the source region, target region, and vehicle eligibility remain unchanged. Within each physically connected scenario, the same $\mathsf{scr} \times \mathsf{opn}$ four-condition counterfactual design is retained, with physical connectivity held fixed as a common scenario-level condition rather than introduced as a third experimental factor. Apart from the two manipulated factors, the road and regional definitions, initial traffic conditions, vehicle-behaviour and baseline control rules, risk measurement, message rules, and pairing of stochastic processes are held consistent according to the pre-specified design. Structural physical connectivity indicates only that ordinary physical propagation from the source region to the target region is possible; it does not imply that a disturbance originating from the specified source-risk event has actually propagated to the target region or produced an actual vehicle response there. Such a response must still be established using the temporally ordered valid physical-propagation chains defined above.

Under physically connected conditions, Effect and Attribution are not redefined but are evaluated using the same counterfactual effect measures and attribution criteria established above. The target-region Effect continues to be assessed through the four-condition interaction contrast, while Attribution continues to require a traceable relationship between the observed risk difference, the specified source-risk

event, and the corresponding information-action pathway. These analyses and the subsequent temporal analyses are all conducted according to pre-specified rules and are not selectively performed on the basis of whether any preceding result reaches statistical significance. Information-first timing can therefore contribute behavioural-level temporal evidence for information-mediated action at a distance only when both Effect and Attribution are also supported; the corresponding risk-outcome-level temporal-precedence requirement is evaluated separately below.

The primary test of Temporal precedence is conducted within condition $(\mathsf{scr}, \mathsf{opn}) = (1,1)$ (Fig. 2c), because this condition simultaneously allows the specified source-risk event to trigger the remote-information pathway and preserves the possibility that ordinary physical propagation from the source produces an actual vehicle response in the target region. For experimental units satisfying the pre-specified quality-control and temporal-analysis requirements, the earliest information-response time $t_\omega^{\mathsf{info\text{-}resp}}$ and the earliest physical-response time $t_\omega^{\mathsf{phy\text{-}resp}}$ are both extracted from the same $(1,1)$ run and are referenced to the same specified source-risk event $\eta_\omega^{\mathsf{src}}$. When both times are finite, they are compared directly as a matched pair. If no qualifying actual response attributable to the information pathway or to ordinary physical propagation is observed within the pre-specified observation range, the corresponding response time is retained as unobserved or censored and handled according to the pre-specified statistical procedure, rather than being imputed post hoc or selectively excluded to construct a temporal ordering.

Condition $(1,0)$ provides a paired reference for the timing of ordinary physical propagation in the absence of remote-information influence and is used for sensitivity analysis to examine whether the remote-information pathway materially alters the subsequent timing of the physical response. It does not, however, replace the primary within-run comparison in condition $(1,1)$, where the information and physical pathways are both active and their actual competition can be evaluated directly.

For experimental units with comparable response times, information-first is determined using the strict criterion defined above,

$$t_\omega^{\mathsf{info\text{-}resp}} < t_\omega^{\mathsf{phy\text{-}resp}}, \tag{35}$$

and the corresponding temporal advantage of the specified remote information over ordinary physical propagation is quantified by

$$\Delta t_\omega^{\mathsf{lead}} = t_\omega^{\mathsf{phy\text{-}resp}} - t_\omega^{\mathsf{info\text{-}resp}}. \tag{36}$$

A positive $\Delta t_\omega^{\mathsf{lead}}$ indicates that the information pathway produces an actual vehicle response in the target region before ordinary physical propagation from the same source-risk event does so. This criterion establishes Temporal precedence at the behavioural-response level only. An earlier vehicle response attributable to remote information does not by itself demonstrate that aggregate target-region risk has already changed, nor does it establish that the regional risk effect emerged before ordinary physical propagation began to influence risk formation in the target region.

To further test Temporal precedence at the risk-outcome level, we use the pre-specified early risk-evaluation window $\mathcal{T}_\omega^{\mathsf{prephy}}$ within the same physically connected four-condition experimental unit. Target-region risk is calculated separately for each of the four counterfactual runs over this window as $R_\omega^{\mathsf{tgt,prephy}}(s,o)$, and the corresponding early four-condition interaction effect for experimental unit ω is defined as

$$\begin{aligned}\delta_\omega^{\mathsf{NL,prephy}} = &\left[R_\omega^{\mathsf{tgt,prephy}}(1,1) - R_\omega^{\mathsf{tgt,prephy}}(1,0)\right] \\ &- \left[R_\omega^{\mathsf{tgt,prephy}}(0,1) - R_\omega^{\mathsf{tgt,prephy}}(0,0)\right].\end{aligned} \tag{37}$$

This quantity uses exactly the same four-condition matched structure and risk definition as the primary effect $\delta_\omega^{\mathsf{NL}}$, differing only in that risk evaluation is restricted to the pre-specified window $\mathcal{T}_\omega^{\mathsf{prephy}}$. Accordingly, $\delta_\omega^{\mathsf{NL,prephy}}$ is used to test whether the target-region risk effect associated with the specified remote information is already detectable before ordinary physical propagation from the source begins to influence risk formation in the target region; it does not replace the primary Effect defined over the full core outcome window.

Whether the early risk-evaluation window supports this interpretation must also be verified using the physical-propagation tracking rules locked before Stage I. Specifically, in the source-risk-treatment conditions $(1,0)$ and $(1,1)$, if ordinary physical propagation originating from the same specified source-risk event is found to have altered the vehicle-interaction states contributing to the target-region risk calculation within $\mathcal{T}_\omega^{\mathsf{prephy}}$, then that experimental unit is not considered to satisfy Temporal precedence at the risk-outcome level. In such cases, $\mathcal{T}_\omega^{\mathsf{prephy}}$ is not shortened or shifted post hoc on the basis of the physical-propagation time observed in the formal run, nor is the affected run selectively removed and the early four-condition comparison reconstructed from the remaining conditions. Instead, the case is retained and reported according to the pre-specified temporal-evaluation rules. The primary Effect, Attribution, and behavioural-level information-first status of that experimental unit are still evaluated independently according to their respective criteria, but the unit itself does not provide supporting evidence for risk-outcome-level Temporal precedence.

Taken together, the physically connected setting jointly evaluates four complementary forms of evidence: the remote-information Effect estimated over the full core outcome window; Attribution consistent with the specified message and its traceable information-action pathway; behavioural-level Temporal precedence characterised by

$$t_\omega^{\mathsf{info\text{-}resp}} < t_\omega^{\mathsf{phy\text{-}resp}};$$

and risk-outcome-level Temporal precedence supported jointly by $\delta_\omega^{\mathsf{NL,prephy}}$ and the validity of the pre-specified early risk-evaluation window. These outcomes are summarised across the pre-specified population of experimental units, together with the corresponding effect sizes, occurrence proportions, distributions of information lead time, and statistical uncertainty. Whether the evidence satisfies the core criteria for supporting information-mediated action at a distance is determined using pre-specified decision rules rather than any single experimental unit or individual statistical result. Subsequent analyses then vary traffic states, spatial conditions, and communication conditions to determine the range over which the effect persists and to identify its failure boundaries.

Stage II: Safety-constrained control using remote risk information

The study proceeds to Stage II only if the core Stage I evidence for information-mediated action at a distance satisfies the pre-specified criteria for Stage II entry. Stage II does not re-test whether the remote risk-information effect exists, nor is control performance treated as additional evidence for the Effect, Attribution, or Temporal precedence established in Stage I. Instead, Stage II examines whether the remote risk information identified in Stage I can serve as an effective decision input for vehicle control and improve target-region risk while satisfying pre-specified safety and operational constraints. Stage I and Stage II therefore address two distinct questions: whether the information effect can be reliably identified, and whether the identified effect can be usefully exploited by vehicle control. Accordingly, whether the subsequent control experiments yield risk improvement, no appreciable improvement, or improvements accompanied by efficiency or ride-comfort costs does not retrospectively alter the Stage I identification conclusions.

Stage II uses control experimental units that are independent of the Stage I four-condition experimental units ω, denoted by κ. To avoid using the same experimental outcomes for both policy development and final performance evaluation, all Stage II control experimental units are partitioned, before policy development begins, according to pre-specified scenario-stratification and randomisation rules into a development set $\mathcal{K}^{\mathsf{dev}}$ and an independently held-out final-evaluation set $\mathcal{K}^{\mathsf{eval}}$, such that

$$\mathcal{K} = \mathcal{K}^{\mathsf{dev}} \dot{\cup} \mathcal{K}^{\mathsf{eval}}, \qquad \mathcal{K}^{\mathsf{dev}} \cap \mathcal{K}^{\mathsf{eval}} = \varnothing. \tag{38}$$

The development set $\mathcal{K}^{\mathsf{dev}}$ is used exclusively for control-policy development, hyperparameter determination, candidate-policy comparison, and Pareto-based policy screening. The held-out set $\mathcal{K}^{\mathsf{eval}}$ does not participate in policy selection before the development procedure has been completed and frozen, and its results are used only for independent final evaluation of the frozen candidate policies. Apart from experimental-unit identity and the corresponding stochastic realisations, both sets are generated according to the scenario-generation rules, traffic-condition ranges, communication conditions, and evaluation protocol pre-specified before formal Stage II experiments. If separate experiments are subsequently introduced specifically to assess extrapolation or failure boundaries, their role is distinguished from the development–evaluation split used here to prevent information leakage between policy development and final evaluation.

For any $\kappa \in \mathcal{K}$, we retain the definitions of the target-vehicle set and the set of vehicles eligible to receive and adopt remote messages introduced above, denoted by $\mathcal{V}_{\kappa}^{\mathsf{tgt}}$ and $\mathcal{V}_{\kappa}^{\mathsf{msg}}$, respectively. We further define the subset of vehicles that are also eligible for direct longitudinal control as

$$\mathcal{V}_{\kappa}^{\mathsf{ctrl}} \subseteq \mathcal{V}_{\kappa}^{\mathsf{msg}} \subseteq \mathcal{V}_{\kappa}^{\mathsf{tgt}}. \tag{39}$$

Only vehicles in $\mathcal{V}_{\kappa}^{\mathsf{ctrl}}$ directly receive control outputs from the Stage II policy. Other target vehicles do not receive direct control requests from this policy, although their motion states and risk still contribute to the aggregate evaluation of the target region. For a controlled vehicle $v \in \mathcal{V}_{\kappa}^{\mathsf{ctrl}}$, let $\mathcal{I}_{v,t}^{\mathsf{loc}}$ denote the local longitudinal traffic information available to the vehicle at time t. In the longitudinal-control setting considered here, this information includes the vehicle's own longitudinal speed $u_{v,t}$ and actual longitudinal acceleration $a_{v,t}$, as well as, when a valid leader is present, the net longitudinal gap $g_t^{(v'\to v)}$ and closing speed $\Delta u_t^{(v'\to v)}$.

For candidate control policies that are permitted to use remote risk information, let $\mathcal{I}_{v,t}^{\mathsf{ctrl}}$ denote the complete set of information actually available to vehicle v for control decision-making at time t. This set consists of the local information $\mathcal{I}_{v,t}^{\mathsf{loc}}$ together with the pre-specified decision-relevant payloads of specified risk messages that have been adopted by time t; messages that have not yet been delivered or adopted, future traffic states, and post hoc outcomes are not included in the current control information. Let π denote a longitudinal control policy. The requested longitudinal control input is then generated as

$$x_{v,t}^{\mathsf{cmd}} = \pi\big(\mathcal{I}_{v,t}^{\mathsf{ctrl}}\big). \tag{40}$$

In the longitudinal-control setting considered here, $x_{v,t}^{\mathsf{cmd}}$ specifies the longitudinal acceleration requested by the policy and is not equivalent to the actual longitudinal acceleration $a_{v,t}$ subsequently observed for the vehicle. The requested control is processed through the pre-specified safety handling and actuator mapping to produce the actually executed control $x_{v,t}^{\mathsf{act}}$, which then affects the subsequent vehicle motion through the vehicle dynamics, including actual acceleration, speed, and inter-vehicle state. Stage II therefore optimises the decision rule mapping the information actually available to the vehicle to a longitudinal control request, while preserving the distinction among local information, adopted remote information, requested control, actually executed control, and the resulting physical vehicle response.

Under the control-policy definition above, not every decision rule mapping $\mathcal{I}_{v,t}^{\mathsf{ctrl}}$ to $x_{v,t}^{\mathsf{cmd}}$ is admissible for subsequent optimisation. Stage II does not modify the pre-specified longitudinal vehicle dynamics or actuator-update rules of the simulation environment; instead, the requested control generated by a policy is passed through this fixed execution process to produce the actual control and subsequent vehicle motion, subject to the following hard constraints. First, the requested longitudinal control input must remain within the pre-specified control-interface bounds,

$$\underline{x}^{\mathsf{cmd}} \le x_{v,t}^{\mathsf{cmd}} \le \overline{x}^{\mathsf{cmd}}, \tag{41}$$

where $\underline{x}^{\mathsf{cmd}}$ and $\overline{x}^{\mathsf{cmd}}$ denote the lower and upper bounds, respectively, on the longitudinal control request; Eq. (41) excludes requests outside the permitted control-interface range. After safety handling and actuator action, the vehicle's actual longitudinal acceleration must further satisfy

$$\underline{a} \le a_{v,t} \le \overline{a}, \tag{42}$$

where $\underline{a}$ and $\overline{a}$ denote the allowable lower and upper bounds on actual longitudinal acceleration, thereby constraining the range of physical longitudinal responses that the vehicle can produce. To further limit excessively abrupt changes between consecutive control instants, the actual longitudinal jerk of vehicle v at time t is defined as

$$j_{v,t} = \frac{a_{v,t} - a_{v,t-\Delta t}}{\Delta t},$$

and must satisfy the pre-specified longitudinal jerk bounds

$$\underline{j} \le j_{v,t} \le \overline{j}, \tag{43}$$

where $\underline{j}$ and $\overline{j}$ are the allowable lower and upper jerk bounds. This constraint specifies the basic range of permissible control-action variation, whereas the comfort objective defined below is used to compare policies that already satisfy this hard bound. In addition to these execution constraints, a controlled vehicle and its valid leader must satisfy the pre-specified longitudinal safety condition

$$g_t^{(v'\to v)} \ge g^{\mathsf{safe}}\Big(u_{v,t}, \Delta u_t^{(v'\to v)}\Big), \tag{44}$$

where $g^{\mathsf{safe}}(\cdot)$ denotes the minimum allowable safety gap determined from the current longitudinal motion state. Equation (44) prevents a policy from obtaining apparent performance gains by entering longitudinal safety states that are prohibited a priori. Finally, remote risk information may enter the current control decision only after it has been adopted by the target vehicle,

$$q \in \mathcal{I}_{v,t}^{\mathsf{ctrl}} \quad \Longrightarrow \quad t_{q,v}^{\mathsf{adopt}} \le t, \tag{45}$$

where adoption retains the definition introduced above: the message has been accepted by the vehicle's decision or control module as a valid decision input, rather than merely having been delivered or being eligible for adoption. Together, the control bounds, safety conditions, information-availability rules, and fixed vehicle-execution mechanism define the policy space that may be searched and evaluated in Stage II,

$$\begin{aligned}\Pi^{\mathsf{adm}} = \Big\{\pi : \pi \text{ complies with the pre-specified} \\ \text{control-interface, information-availability,} \\ \text{constraint-handling, and fixed} \\ \text{vehicle-execution rules}\Big\}.\end{aligned} \tag{46}$$

Thus, Π^{adm} denotes the candidate-policy space that is admissible at the levels of policy structure and execution protocol; admissibility is not determined retrospectively by inspecting outcomes in $\mathcal{K}^{\mathsf{eval}}$.

Hard-constraint violations that may nevertheless occur during actual execution are handled separately in the development and final-evaluation sets according to pre-specified feasibility rules. Violations observed in $\mathcal{K}^{\mathsf{dev}}$ may be used during policy development to eliminate or revise candidate strategies. Once the policies to be evaluated have been frozen, however, any hard-constraint violation occurring in $\mathcal{K}^{\mathsf{eval}}$ is retained in full as part of the final feasibility outcome and may not be used to retune the policy and repeat the evaluation.

Within the admissible policy space Π^{adm}, Stage II evaluates control policies along three distinct dimensions: target-region risk, traffic efficiency, and longitudinal ride comfort of directly controlled vehicles. The risk objective serves a different role from the longitudinal safety constraints defined above: the safety constraints specify states that a policy is not permitted to enter, whereas the risk objective further distinguishes the aggregate target-region risk produced by different policies that satisfy the pre-specified hard constraints and feasibility rules. To maintain consistency with Stage I, Stage II uses the same risk-measurement, target-vehicle eligibility, and regional-aggregation rules defined above, together with a fixed outcome window referenced to the specified source-risk event. For control experimental unit κ, let the corresponding evaluation window be denoted by $\mathcal{T}_\kappa^{\mathsf{out}}$. The risk objective of policy π in experimental unit κ is defined as

$$J_\kappa^{\mathsf{risk}}(\pi) = R_\kappa^{\mathsf{tgt}}(\pi), \tag{47}$$

where $R_\kappa^{\mathsf{tgt}}(\pi)$ denotes the aggregate target-region risk evaluated over $\mathcal{T}_\kappa^{\mathsf{out}}$ using the common risk rules defined above. A smaller $J_\kappa^{\mathsf{risk}}(\pi)$ therefore indicates lower target-region risk.

Traffic efficiency is treated as a second evaluation dimension, distinct from target-region risk and control comfort. For control experimental unit κ and policy π, let the efficiency objective evaluated according to the pre-specified efficiency-assessment rules be denoted by

$$J_\kappa^{\mathsf{eff}}(\pi).$$

The specific quantity used to measure traffic efficiency and its aggregation form will be determined before formal Stage II experiments according to the control scenario, target-region definition, and evaluation requirements, and will be incorporated into the Stage II Protocol Lock. Candidate measures may include vehicle operating speed within the target region, travel time, delay, or other quantities that characterise traffic operational performance; however, none of these candidate measures is treated as the final efficiency definition adopted in this study before completion of the Protocol Lock.

To maintain a common minimisation convention in the subsequent multi-objective formulation, the final efficiency measure is transformed, according to a pre-specified rule, so that a smaller value of $J_\kappa^{\mathsf{eff}}(\pi)$ always corresponds to better traffic efficiency. If the ultimately selected raw efficiency measure is benefit-oriented, a pre-specified monotonic transformation is applied to reverse its direction; if it is already cost-oriented, it is incorporated directly according to the corresponding locked rule. The definition of the efficiency measure, its directional transformation, the evaluated population, and the aggregation procedure are held fixed across all candidate policies and matched experimental runs. None of these elements may be selected or switched post hoc on the basis of the magnitude of risk improvement, a policy's Pareto position, or the final experimental outcomes.

Ride comfort is evaluated using the magnitude of the actual longitudinal jerk of directly controlled vehicles. Only controlled vehicle–time observations for which the vehicle remains within the target region at both the current and immediately preceding time steps, and therefore has a valid consecutive state pair, are included in the calculation. We define

$$J_\kappa^{\mathsf{comf}}(\pi) = \underset{\substack{t\in\mathcal{T}_\kappa^{\mathsf{out}},\, v\in\mathcal{V}_\kappa^{\mathsf{ctrl}} \\ I_{\mathcal{R}^{\mathsf{tgt}}}(v,t)=1,\, I_{\mathcal{R}^{\mathsf{tgt}}}(v,t-\Delta t)=1}}{\mathrm{mean}} j_{v,t}^2(\pi). \tag{48}$$

Thus, control smoothness is characterised by the mean squared actual longitudinal jerk over valid observations within the evaluation window. A smaller $J_\kappa^{\mathsf{comf}}(\pi)$ indicates smoother changes in the actual longitudinal acceleration of directly controlled vehicles. This objective is used to distinguish comfort among candidate policies that have already satisfied the non-violable jerk bounds specified in Eq. (43).

During policy development, the experimental-unit-level risk, efficiency, and comfort outcomes are aggregated into

policy-level objectives using only the development set $\mathcal{K}^{\mathsf{dev}}$. Because formal Stage II experiments may include different traffic states, scenario types, or experimental strata, the aggregation across experimental units is not assumed a priori to be a simple unweighted mean. Let $\mathcal{A}_{\mathcal{K}^{\mathsf{dev}}}$ denote the development-set aggregation operator to be specified and frozen before formal Stage II experiments. For $d \in \{\mathsf{risk}, \mathsf{eff}, \mathsf{comf}\}$, the overall objective of policy π in evaluation dimension d over the development set is defined as

$$J^{d,\mathsf{dev}}(\pi) = \mathcal{A}_{\mathcal{K}^{\mathsf{dev}}}\left(\left\{J^{d}_{\kappa}(\pi) : \kappa \in \mathcal{K}^{\mathsf{dev}}\right\}\right). \tag{49}$$

The three-objective evaluation vector used during policy development is therefore

$$\boldsymbol{J}^{\mathsf{dev}}(\pi) = \left(J^{\mathsf{risk},\mathsf{dev}}(\pi), J^{\mathsf{eff},\mathsf{dev}}(\pi), J^{\mathsf{comf},\mathsf{dev}}(\pi)\right). \tag{50}$$

The specific form of $\mathcal{A}_{\mathcal{K}^{\mathsf{dev}}}$ is determined according to the structure of the Stage II scenarios and the composition of the experimental units before formal Stage II experiments, and is incorporated into the Protocol Lock. If all development experimental units represent the same pre-specified population and carry equal evaluation weight, the aggregation rule may reduce to a simple unweighted mean. If the formal experimental design instead includes units representing different traffic states or scenario strata, a pre-specified stratified aggregation scheme or scenario-weighting rule may be used. Regardless of the form adopted, the corresponding experimental-unit weights, scenario strata, and aggregation rules are fixed in advance and are not adjusted post hoc according to the magnitude of risk improvement achieved by a policy, its Pareto position, or subsequent final-evaluation results. The same scenario-weight structure is also maintained across the risk, efficiency, and comfort objectives.

We do not assign a priori exchange weights among the three objectives of risk, efficiency, and comfort, nor do we collapse them into a single weighted performance score. Policy development and Pareto screening are conducted exclusively on $\mathcal{K}^{\mathsf{dev}}$ by solving

$$\underset{\pi \in \Pi^{\mathsf{adm}}}{\text{minimize}} \quad \boldsymbol{J}^{\mathsf{dev}}(\pi), \tag{51}$$

and the non-dominated policies on the development set are identified according to Pareto dominance. A candidate policy entering the Pareto comparison is said to dominate another if it is no worse in all three development-set objectives of risk, efficiency, and comfort and is strictly better in at least one of them. The candidate policy or candidate-policy set carried forward to independent final evaluation is selected solely on the basis of results from $\mathcal{K}^{\mathsf{dev}}$ and the pre-specified screening rules, and is frozen before any outcomes from $\mathcal{K}^{\mathsf{eval}}$ are accessed.

After policy development has been completed and the candidate policies for evaluation have been frozen, Stage II conducts the final paired evaluation on the independently held-out set $\mathcal{K}^{\mathsf{eval}}$. For each $\kappa \in \mathcal{K}^{\mathsf{eval}}$ and each frozen candidate remote-information control policy $\pi \in \Pi^{\mathsf{adm}}$, we construct a matched local-information reference policy π^{loc}. This reference policy uses the same control structure, control interface, safety handling, actuator mapping, and hard constraints as the candidate policy, but bases its control decisions only on the local longitudinal traffic information available to the vehicle at that time, $\mathcal{I}^{\mathsf{loc}}_{v,t}$, such that

$$x^{\mathsf{cmd},\mathsf{loc}}_{v,t} = \pi^{\mathsf{loc}}\left(\mathcal{I}^{\mathsf{loc}}_{v,t}\right). \tag{52}$$

Accordingly, π^{loc} does not use any decision-relevant payload contained in the specified remote risk messages.

The candidate-policy run and its matched local-information reference run use the same message-generation rules, communication configuration, transmission and delivery mechanisms, with the corresponding stochastic processes paired so that the remote-message communication process follows the same experimental protocol in both runs. The key difference between the two policies lies at the interface through which messages enter vehicle control decisions. Under the candidate remote-information policy, messages satisfying the adoption rules may enter $\mathcal{I}^{\mathsf{ctrl}}_{v,t}$ and contribute to the current control decision. Under π^{loc}, by contrast, remote messages retain their generation, transmission, and delivery records but are not accepted as valid control inputs; they therefore do not constitute actual adoption as defined in this study and do not enter $\mathcal{I}^{\mathsf{loc}}_{v,t}$. The core experimental difference between the two policies is thus whether the specified remote risk information is allowed to enter the vehicle's control decision, rather than whether the communication channel exists or whether the message has an opportunity to reach the vehicle.

The candidate-policy run and its matched local-information reference run are initiated from independently initialised but mutually matched system states. The road and regional definitions, initial traffic conditions, source-risk conditions, vehicle eligibility, background traffic, communication configuration, control-update frequency, evaluation window, and corresponding stochastic processes are matched according to the pre-specified protocol. After the experimental difference takes effect, divergence in vehicle trajectories, subsequent message states, and risk outcomes arising from whether remote information enters the control decision is treated as part of the policy effect and is not required to remain matched.

For each frozen candidate policy, Stage II evaluates $J^{\mathsf{risk}}_{\kappa}$, $J^{\mathsf{eff}}_{\kappa}$, and $J^{\mathsf{comf}}_{\kappa}$ over all experimental units in $\mathcal{K}^{\mathsf{eval}}$ using the common rules defined above. If a candidate policy violates any pre-specified hard constraint in a final-evaluation run, the violation is retained in full and incorporated into the final assessment according to the frozen feasibility rules. Failed runs are not discarded, the policy is not retuned and re-evaluated on the same evaluation unit, and evaluation outcomes are not used to return to the development stage and modify the policy. For candidate policies that do not exhibit unacceptable constraint violations, we further report the paired differences in risk, efficiency, and comfort relative to the matched local-information reference policy π^{loc}, together with the between-unit variability and corresponding statistical uncertainty.

Scenario-level aggregation in the final evaluation follows the scenario strata, weight structure, and aggregation principles frozen before any outcomes from $\mathcal{K}^{\mathsf{eval}}$ are examined. For $d \in \{\mathsf{risk}, \mathsf{eff}, \mathsf{comf}\}$, the overall final-evaluation result of candidate policy π in dimension d is denoted by

$$J^{d,\mathsf{eval}}(\pi) = \mathcal{A}_{\mathcal{K}^{\mathsf{eval}}}\left(\left\{J^{d}_{\kappa}(\pi) : \kappa \in \mathcal{K}^{\mathsf{eval}}\right\}\right), \tag{53}$$

with the corresponding three-objective final-evaluation vector defined as

$$\boldsymbol{J}^{\mathsf{eval}}(\pi) = \left(J^{\mathsf{risk},\mathsf{eval}}(\pi), J^{\mathsf{eff},\mathsf{eval}}(\pi), J^{\mathsf{comf},\mathsf{eval}}(\pi)\right). \tag{54}$$

Here, $\mathcal{A}_{\mathcal{K}^{\text{eval}}}$ follows the same pre-specified scenario-aggregation principles used during development, but is applied only to evaluation experimental units that have not participated in policy development or selection.

Accordingly, the Pareto position of a candidate policy on $\mathcal{K}^{\text{dev}}$ is used only for policy development and candidate screening, whereas Stage II conclusions regarding final control performance, the risk–efficiency–comfort trade-off, and performance improvement relative to the local-information reference policy are based primarily on the independent evaluation results obtained from $\mathcal{K}^{\text{eval}}$. Final evaluation does not require any single policy to achieve a unique absolute optimum across all three objectives. Instead, the risk, efficiency, and comfort outcomes of the frozen candidate policies on the independent evaluation set, together with their statistical uncertainty, are used to determine what reproducible control benefits remote risk information can provide and what corresponding costs or trade-offs accompany those benefits. If the independent evaluation yields no feasible or stable control improvement, this result constrains only the practical exploitability of the identified information effect under the current control design and experimental conditions; it does not retrospectively invalidate the Effect, Attribution, or Temporal precedence established in Stage I.

Calibration, validation and Protocol Lock

Statistical analysis and uncertainty assessment

Data availability

Code availability

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Competing interests

The author declares no competing interests.

Figure legends

Tables